ECEN-601: Homework 4

Due on October 21, 2025 at 11:59 pm

Instructor: Dr. Jean-François Chamberland

Brody Jordan
431005357

Problem 1

Let

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -1 & 0 & 3 & 5 \\ 1 & -2 & 1 & 1 \end{bmatrix}.$$

Find a row-reduced echelon matrix R which is row-equivalent to A and an invertible 3×3 matrix P such that $R = PA$.

Solution

Problem 2

Let V be the set of all complex-valued functions f on the real line such that (for all $t \in \mathbb{R}$)

$$f(-t) = \overline{f(t)}.$$

The bar denotes complex conjugation. Show that V, with the operations

$$(f+g)(t) = f(t) + g(t)$$
$$(cf)(t) = cf(t)$$

is a vector space over the field of real numbers. Give an example of a function in V which is not real-valued.

Solution

Problem 3

Let F be a field and let n be a positive integer ($n \geq 2$). Let V be the vector space of all $n \times n$ matrices over F. Which of the following sets of matrices A are subspaces of V?

(a) all invertible A

(b) all non-invertible A

(c) all A such that $AB = BA$, where B is some fixed matrix in V

(d) all A such that $A^2 = A$

Solution

Problem 4

Let W_1 and W_2 be subspaces of a vector space V such that the set-theoretic union of W_1 and W_2 is also a subspace. Prove that one of the spaces W_i is contained in the other.

Solution

Problem 5

Let W_1 and W_2 be subspaces of a vector space V such that $W_1+W_2 = V$ and $W_1 \cap W_2 = \{\underline{0}\}$. Prove that for each vector α in V there are unique vectors α_1 in W_1 and α_2 in W_2 such that $\alpha = \alpha_1 + \alpha_2$.

Note: If $S_1, S_2, \ldots, S_k$ are subsets of a vector space V, the set of all sums

$$\alpha_1 + \alpha_2 + \cdots + \alpha_k$$

or vectors α_i in S_i is called the sum of the subsets $S_1, S_2, \ldots, S_k$ and is denoted by

$$S_1 + S_2 + \cdots + S_k$$

of by

$$\sum_{i=1}^{k} S_i.$$

Solution

Problem 6

Let V be a vector space over the complex numbers. Suppose that α, β, and γ are linearly independent vectors in V. Prove that $(\alpha + \beta)$, $(\beta + \gamma)$, and $(\gamma + \alpha)$ are linearly independent.

Solution